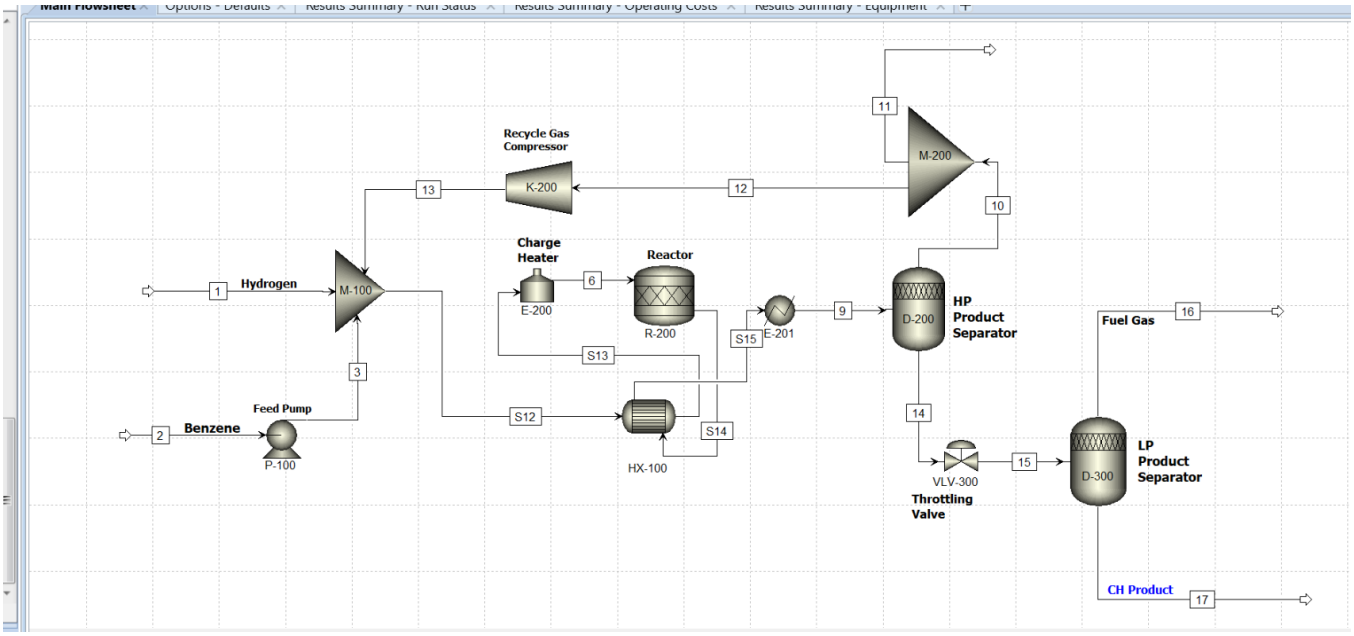
ASPEN PLUS V15 / CHEMICAL ENGINEERING CASE STUDY

Cyclohexane production and energy optimization

Benzene hydrogenation | Process simulation | Heat recovery | Utility economics

53.35%	1.778 MW	$15.44/h
Modeled reduction in total utility cost	Heat recovered in HX-100	Modeled gross utility savings



Heat-recovery flowsheet built in Aspen Plus. HX-100 connects the hot reactor outlet with the cooler feed stream; see Page 4 for a detailed explanation.

Parthav Ajaykumar Patel | University of Illinois Chicago

Academic case study | CHE 433 | Fall 2026

Purpose: evaluate whether process-to-process heat recovery can reduce external heating and cooling while preserving the modeled process targets.

01 / CONTEXT AND OBJECTIVES

What is the product, and why optimize the process?

Cyclohexane (C6H12) is a six-carbon-ring compound used as an industrial chemical intermediate, particularly in supply chains that lead to nylon precursors. One established production route is hydrogenation of benzene.

Main reaction: C6H6 + 3 H2 → C6H12

Benzene reacts with hydrogen to form cyclohexane. The model includes feed preparation, the reactor, cooling, gas/liquid separation, a purge to manage nonreacting gases, and recycle of useful hydrogen.

The engineering question

The incoming reactor feed requires heating, while the reactor outlet requires cooling. Can a heat exchanger transfer energy directly between these two streams and lower the purchased utility duty?

Objective	How it was evaluated
Create a baseline	Run the Aspen Plus benzene-hydrogenation flowsheet at the assigned operating specifications.
Maintain process targets	Check the H2-to-benzene ratio and combined methane/nitrogen target after recycle convergence.
Add heat recovery	Insert HX-100 between the hot effluent and the colder mixed feed.
Compare on one basis	Use common utility tariffs and compare duties, hourly costs, and convergence.

Scope and interpretation

This report documents a steady-state academic simulation. The 99.90% benzene conversion is a specified reactor-model input, not a conversion independently predicted from reaction kinetics. Plant design, product certification, heat-exchanger procurement, and a full economic feasibility study were not completed.

How the cyclohexane process works

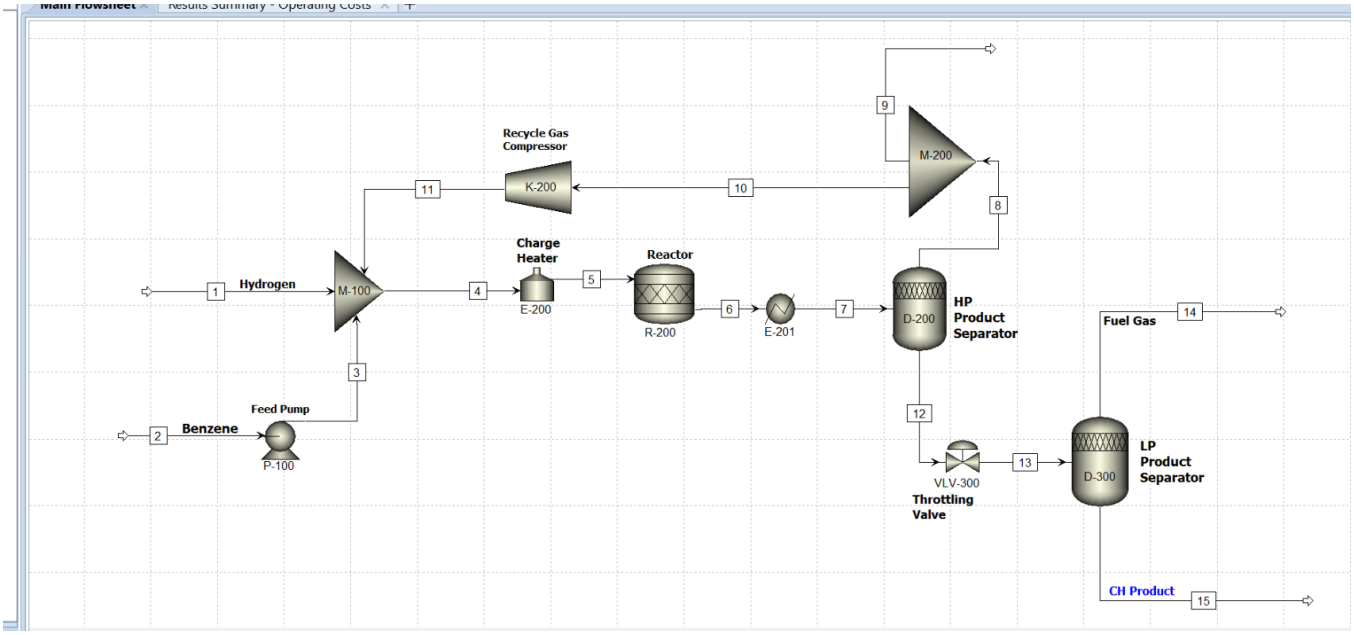


Figure 1. Original Aspen Plus base-case process flowsheet, before HX-100 heat recovery. The feed heater and effluent cooler rely on external utilities.

Equipment / step	What happens and why
P-100, M-100	Pump benzene to process pressure and mix with fresh/recycled hydrogen-bearing gas.
E-200, R-200	Heat the mixture and react benzene with hydrogen to form cyclohexane.
E-201, D-200	Cool reactor effluent and separate the gas-rich and liquid-rich streams.
VLV-300, D-300	Lower pressure and separate additional flashed vapor to recover product-rich liquid.
Purge, K-200	Bleed a portion of gas to limit inert buildup; compress and recycle useful gas.

Operating and convergence checks

The specified reactor temperature was 400 °F (204.4 °C) with 99.90% benzene conversion. The simulation converged at H2/benzene molar ratio 5.00005 (target 5.0) and a combined CH4 + N2 mole fraction 0.08003 (target 0.08). The source summary did not explicitly identify the stream location of the latter target; it is therefore reported only as a design-specification check.

03 / DESIGN CHANGE

Putting the hot outlet to work

The hot reactor outlet and the incoming feed are a useful heat-transfer pair. In the revised flowsheet, HX-100 transfers energy across a heat-exchanger wall: the hot process stream cools, and the cold process stream warms, without mixing.

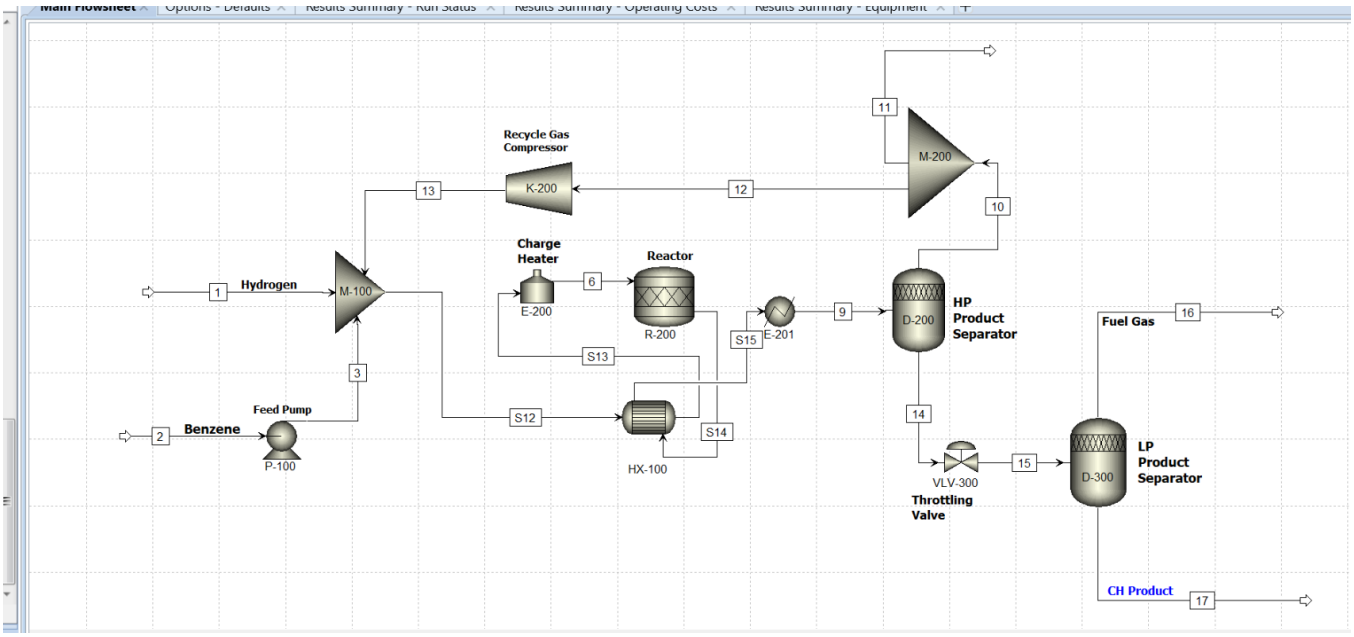


Figure 2. Aspen Plus heat-recovery case. HX-100 is installed between the hot reactor effluent and the colder mixed feed.

Step	Engineering meaning
1 / Match streams	Identify a process stream needing cooling and another requiring heating.
2 / Transfer heat	Use HX-100 to reduce both the feed-heater and downstream-cooler duties.
3 / Re-converge	The modified recycle loop required raising maximum iterations from 30 to 100.
4 / Verify	Compare heat duties and utility costs at the same utility-price basis.

A heat exchanger redistributes energy; it does not create energy. For an insulated exchanger, hot-side heat lost is approximately equal to cold-side heat gained. This was a direct exchanger match, not a completed plant-wide pinch network study.

04 / VERIFIED SIMULATION RESULTS

What changed after adding HX-100?

Metric	Base case	Heat recovery	Reduction
Heating duty	531,716 cal/s	106,736 cal/s	424,980 cal/s
Cooling duty	2,393,110 cal/s	1,968,130 cal/s	424,980 cal/s
Heating cost	$17.6314/h	$3.53931/h	$14.09209/h
Cooling cost	$7.57473/h	$6.22957/h	$1.34516/h
Electricity cost	$3.72792/h	$3.72784/h	Negligible
Total utility cost	$28.9341/h	$13.4967/h	53.35%

Heat recovery, calculated from heater-duty reduction

531,716 - 106,736 = 424,980 cal/s

424,980 cal/s × 4.184 J/cal = 1,778,116 W ≈ 1.778 MW. Cooling duty falls by the same 424,980 cal/s.

Utility-cost reduction

($28.9341 - $13.4967) / $28.9341 × 100 ≈ 53.35%. The estimated savings are $15.4374 per operating hour. At an illustrative 8,000 hours of operation per year, gross modeled savings are $123,499/year before exchanger capital cost, maintenance, and financing.

How to read the percentages

Heating cost decreases by approximately 79.9%, while cooling cost falls by about 17.8%. The overall 53.35% is a weighted utility-cost change, using the case-study tariffs. It is not a 53.35% reduction in total cyclohexane production cost.

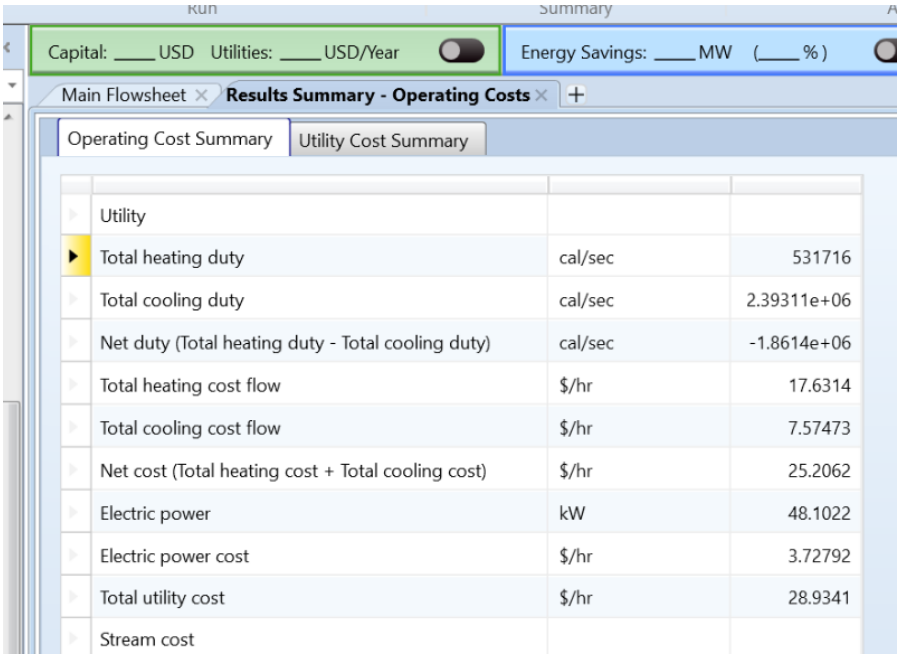
Utility-price basis

10-bar steam: $2.20/GJ; cooling water: $0.21/GJ (30–40 °C); electricity: $0.0775/kWh. Results depend on these assumed prices and operating hours. A real project should test local tariffs and future price sensitivity.

Original Aspen utility outputs

The screenshots below are retained from the model runs. They provide a direct record of the displayed utility costs for each simulated case.

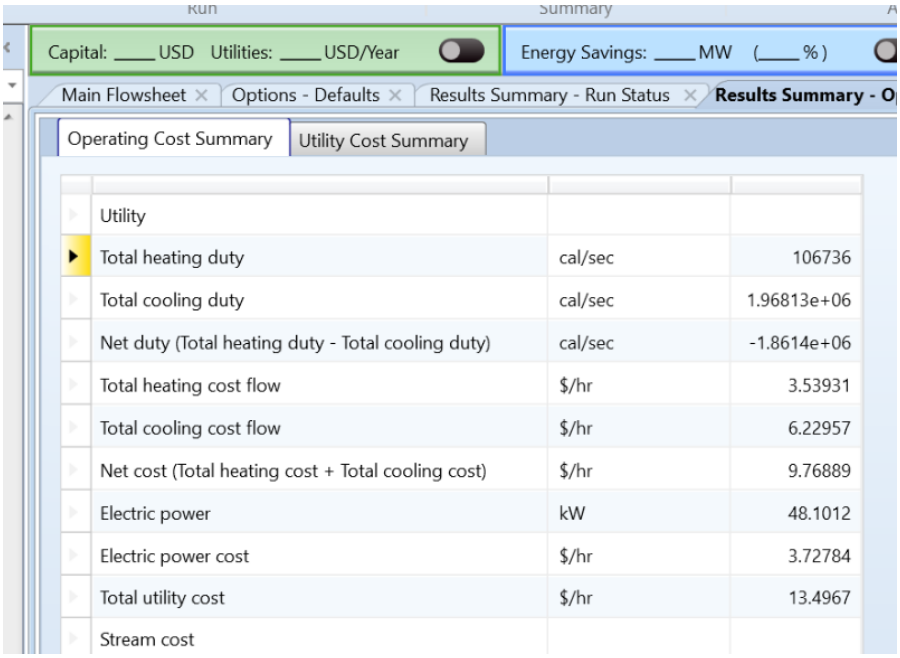
Base-case Aspen operating-cost summary



Utility		
Total heating duty	cal/sec	531716
Total cooling duty	cal/sec	2.39311e+06
Net duty (Total heating duty - Total cooling duty)	cal/sec	-1.8614e+06
Total heating cost flow	$/hr	17.6314
Total cooling cost flow	$/hr	7.57473
Net cost (Total heating cost + Total cooling cost)	$/hr	25.2062
Electric power	kW	48.1022
Electric power cost	$/hr	3.72792
Total utility cost	$/hr	28.9341
Stream cost		

Figure 3. Base-case Aspen output; total utility cost: $28.9341 per hour.

Heat-recovery Aspen operating-cost summary



Utility		
Total heating duty	cal/sec	106736
Total cooling duty	cal/sec	1.96813e+06
Net duty (Total heating duty - Total cooling duty)	cal/sec	-1.8614e+06
Total heating cost flow	$/hr	3.53931
Total cooling cost flow	$/hr	6.22957
Net cost (Total heating cost + Total cooling cost)	$/hr	9.76889
Electric power	kW	48.1012
Electric power cost	$/hr	3.72784
Total utility cost	$/hr	13.4967
Stream cost		

Figure 4. Heat-recovery Aspen output; total utility cost: $13.4967 per hour.

06 / ECONOMICS AND LIMITATIONS

What the numbers prove - and what they do not

The utility saving is promising, but utility savings alone do not establish that an additional exchanger should be purchased. The model does not provide enough verified HX-100 mechanical design data to support an installed cost or payback claim.

Item	Status
Base-case equipment purchase cost	$1,374,900 (APEA estimate)
Base-case installed cost	$2,071,600 (APEA estimate)
Base-case total capital cost	$2,964,935 (APEA estimate; rounded about $2.96 million)
HX-100 installed cost / simple payback	Not calculated; exchanger area, pressure drop, materials and installation costs required.

The APEA values are for the base case only; the reactor was excluded under the assignment scope. They do not represent the installed cost of the optimized HX-100 design.

What an engineer would calculate next

- Thermal feasibility: establish the minimum end-temperature approach and check exchanger effectiveness using actual stream enthalpies and phase behavior.
- Area: estimate with $Q = U A \Delta T_{lm}$ (and a correction factor where applicable).
- Practical constraints: estimate pressure drops, fouling resistance, material compatibility, instrumentation, and safe operability.
- Economics: estimate purchased/installed HX-100 cost, operating changes, and simple payback under several utility-price cases.

Data not available in the verified source summary

Product flow rate and purity, purge flow rate, hydrogen-feed composition, pump/compressor power, HX-100 area, and exchanger approach temperatures are not reproduced here because verified values were not available for this report. These should be exported from Aspen before making a complete plant-performance claim.

07 / REFLECTION AND SOURCES

What I learned from this simulation

Connecting the process. The purge and recycle loop affected the next pass through the mixer. Convergence required attention to the feedback between equipment rather than solving every block in isolation.

Controlling composition. Hydrogen recycle saves useful reactant, but a purge is necessary to prevent methane and nitrogen from building indefinitely. Design specifications helped retain the assignment targets.

Understanding heat exchangers. I learned why hot and cold process streams can be paired, how energy conservation connects their duties, and why recovering heat can reduce both heating and cooling purchases.

Checking the meaning of an optimization. A successful numerical result is only part of engineering: temperature feasibility, equipment sizing, safety, and total economics decide whether a real installation is worthwhile.

Key takeaway

A useful Aspen model is not simply one that converges. It is a model whose design targets, balances, assumptions, and cost comparisons can be explained and checked.

Sources and reproducibility

1. Parthav A. Patel, Aspen Plus Simulation and Utility-Cost Reduction Report, CHE 433 Homework Set 02, University of Illinois Chicago, September 20, 2026. Original Aspen flowsheet and operating-cost screenshots reproduced from the case study.

2. Aspen Plus V15, Peng-Robinson equation-of-state model, steady-state simulation results and Aspen Process Economic Analyzer (APEA) base-case summary as documented in the project.

3. General method background: pinch analysis and heat-exchanger design concepts. Technical methods here are explanatory; no project-wide pinch targets have been claimed.

Disclosure: ChatGPT assisted with model troubleshooting, calculation checks, writing, and report preparation. The simulation data and project case-study results are attributed to the student work referenced above.

Project website: https://parthavpatel12-maker.github.io/cyclohexane-optimization/